INCH-POUND

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MILITARY SPECIFICATION

FILTER ASSEMBLY

This specification is approved for use by the U.S. Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an electromagnetic interference (EMI) filter assembly which provides high and low frequency filtration in the electro-hydraulic stabilization system of the M60A1 tank.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5915

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SPECIFICATIONS

MILITARY

MIL-S-45743 - Soldering, Manual Type, High Reliability, Electrical and Electronic Equipment

STANDARDS

MILITARY

MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-202 - Test Methods of Electronic and Electrical Component Parts.
MIL-STD-220 - Method of Insertion - Loss Measurement.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-810 - Environmental Test Methods.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Printing Services Office, Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

11674230 - Filter Assembly.

(Copies of the drawings should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for related associated detail specifications, specification sheets, or MS sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), sample filters shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Materials. The materials shall be as specified herein and in referenced documents. The materials shall be uniform in quality and free from defects and imperfections that might affect the serviceability and reliability of the finished product. All materials which are not specifically described herein shall be of the highest quality and suitable for the purpose intended (see 4.7.1).

3.2.1 Recycled, virgin, and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 4.7.1).

3.3 Design and construction. Filters shall conform to drawing 11674230 and applicable component drawings and military standards (see 4.7.1 and 4.7.2).

3.3.1 Weight. The weight of the assembly shall not be more than six pounds (see 4.7.2).

3.3.2 Standards of design and manufacture. Unless otherwise specified, the assembly shall conform to the following requirements of MIL-STD-454 (see 4.7.1).

Number	Requirements
4	Fungus-Inert Materials
7	Interchangeability
10	Electrical Connectors
11	Insulating Materials, Electrical
12	Fastener Hardware
20	Hook-up Wire Selection
21	Castings
34	Nomenclature
55	Enclosures

3.3.3 Soldering. Electrical connections shall be soldered per MIL-S-45743 (see 4.7.1).

3.4 Performance. The connector designations shown on the schematic diagram of Drawing 11674230 shall apply to the performance requirements of 3.4.1 through 3.4.4).

3.4.1 Circuit resistance. Circuit direct current (dc) resistance of the assembly shall conform with requirements shown in table I (see 4.7.3.1).

3.4.2 Insulation resistance. With voltage of 50 + 5 volts direct current (vdc) applied, insulation resistance between each pin of connector P1 or J1 and between pins and enclosure shall be not less than 10 megohms. This requirement does not apply to pin P (see 4.7.3.2).

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3.4.3 Low frequency filter. Input capacitance between pin P of connector P1 or J1 and assembly case shall be 2460 (+1800, -250) microfarads at a frequency of 120 + 5 Hertz (Hz). With a positive voltage of 75 (+0, -5) vdc applied to pin P, dc leakage shall not exceed 366 microamperes (A) over a temperature range of -25°F to +77°F, and dc leakage shall not exceed 2.20 milliamperes (ma) for temperatures greater than +77°F to +125°F (see 4.7.3.3).

3.4.4 High frequency filter. Each circuit through the filter shall provide attenuation as specified in table II (see 4.7.3.4).

TABLE I. Circuit resistance.

Test No.	Connector J1 From Pin	Connector P1 To Pin	Ohms Minimum (Min.)	Ohms Maximum (Max.)	Max. Current Amperes
1	A	A	0.0	0.2	8.0
2	B	B	0.0	0.2	8.0
3	C	C	0.0	0.2	8.0
4	D	D	0.0	0.2	8.0
5	E	E	0.0	0.2	8.0
6	F	F	0.0	0.2	8.0
7	G	G	0.0	0.2	2.0
8	H	H	0.0	0.2	2.0
9	J	J	0.0	0.2	8.0
10	K	K	0.0	0.2	8.0
11	L	L	0.0	0.2	2.0
12	M	M	0.0	0.2	8.0
13	N	N	0.0	0.2	8.0
14	P	P	0.0	0.2	15.0
15	R	R	0.0	0.2	2.0
16	S	S	0.0	0.2	8.0

TABLE II. High frequency filter attenuation.

Frequency, megahertz (mHz)	Attenuation Decibel (db) minimum at Zo=50 ohms
20	38
40	48
50	50
100	50
200	50

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3.5 Environmental conditions.

3.5.1 Temperature. The assembly shall meet the performance requirements specified herein at ambient air temperatures ranging from -25 to +125 degrees Fahrenheit (^{o}F). Assembly shall not be damaged by storage conditions ranging from -65 to +160°F (see 4.7.4.1 and 4.7.4.2).

3.5.2 Vibration. The assembly shall be capable of withstanding vibration profile of figure 1 for a period of 80 minutes (including up to 4 resonance dwells of 13-1/3 minutes each) in each of 3 mutually perpendicular axes (see 4.7.4.3).

3.5.3 Shock.

3.5.3.1 Basic intensity. The assembly shall be capable of withstanding exposure to three half sine wave shock pulses of 50 + 5 gravity unit (g) for a duration of 18 + 3 milliseconds (ms) applied in both directions along three mutually perpendicular axes (see 4.7.4.4.1).

3.5.3.2 High intensity. The assembly shall be capable of withstanding three half sine wave shock pulses of 125 + 12.5g for a duration of 0.6 + 0.2 ms along axes specified herein (see 4.7.4.4.2).

3.5.4 Humidity. The assembly shall meet the requirements specified herein during and after exposure to conditions of up to 99 percent relative humidity (see 4.7.4.5).

3.5.5 Waterproofness. The assembly shall be capable of withstanding a differential pressure of 6 pounds square inch (psi) without leakage with the mating electrical connectors or equivalents installed (see 4.7.4.6).

3.6 Protective finish. Final protective finish shall be as specified on drawing 11674230 (see 4.7.2).

3.7 Identification and marking. The assembly shall be identified by number in accordance with MIL-STD-130 and shall include at least the following (see 4.7.2):

a) Military part number.
b) Manufacturer's name or trademark
c) Manufacturer's serial number.
d) Name of part.

3.8 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9 (see 4.7.2 and 4.7.5).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order (6.2), the contractor may use any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
   1. Preproduction inspection (see 4.4.1).
   2. Initial production (see 4.4.2).

b. Quality conformance inspection (QCI) (see 4.5).
   1. Examination (see 4.5.2).
   2. Tests (see 4.5.3).

c. Control tests (see 4.6).

4.3 Test conditions. 1/ Unless otherwise specified, (see 6.2) all tests shall be conducted under the following conditions:

Temperature	73 ± 18°F
Humidity (relative)	50 ± 30 percent
Pressure	28.5(+2.0, -4.5) inches of mercury
Loads	Maximum current per table I

1/ Prior to the performance tests of 4.7.3.1 through 4.7.3.4, perform the waterproofness test of 4.7.4.6 and the thermal run-in and vibration run-in tests of 4.7.5a and 4.7.5b.

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4.4 First article inspection. First article inspection shall be performed on preproduction and initial production samples as specified herein. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply filters that are fully representative of those inspected as a first article sample. Any changes or deviations of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of two filters. Preproduction inspection shall consist of inspection as specified in table III.

4.4.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select two filters from the first ten filters produced under the production contract for initial production inspection. The initial production sample shall be inspected as specified in table III.

4.4.3 First article test sequence. First article tests shall be conducted on the sample in accordance with the test sequence specified in table IV.

4.4.4 First article inspection failure. Deficiencies found during, or as a result of, the first article inspection shall be cause for rejection of the first article sample until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the first article inspection shall be evidence that all items already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government will not accept products until first article inspection is completed to the satisfaction of the Government.

4.5 QCI. QCI shall include the examination of 4.5.2 and the tests of 4.5.3. Noncompliance with any of the specified requirements in sections 3 and 5 shall be cause for rejection of the sample and the inspection lot.

4.5.1 Sampling plan for examination. Unless otherwise specified (see 6.2), the sampling plan for examination as specified herein shall be used. See 6.4 for definitions of sampling inspection terms.

4.5.1.1 Lot formation. An inspection lot shall consist of all filters having the same military part number, from an identifiable production period, from one manufacturer, from one manufacturing location, submitted at the same time for acceptance.

4.5.1.2 Sampling for examination. The sample for QCI examination shall be randomly selected from the inspection lot in accordance with table V. Before sampling may be initiated, the contractor shall 100 percent inspect a minimum of 20 assemblies.

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TABLE III. Classification of inspection.

Title	Requirement	Inspection	First article	QCI		Control test	
				Exam	Test	1/50	1/200
Group A:							
Materials, design, and construction	3.2, 3.2.1, 3.3 and 3.3.2	4.7.1	X				
Examination	3.3, 3.3.1, 3.5 thru 3.7	4.7.2	X	X			
Group B (Performance):							
Circuit resistance	3.4.1	4.7.3.1	X		X		
Insulation resistance	3.4.2	4.7.3.2	X		X		
Low frequency filter	3.4.3	4.7.3.3	X		X		
High frequency filter	3.4.4	4.7.3.4	X		X		
Group C (Environmental):							
High temperature	3.5.1	4.7.4.1	X			X	
Low temperature	3.5.1	4.7.4.2	X			X	
Vibration	3.5.2	4.7.4.3	X			X	
Shock-Basic intensity	3.5.3.1	4.7.4.4.1	X			X	
Shock-High intensity	3.5.3.2	4.7.4.4.2	X			X	
Humidity	3.5.4	4.7.4.5	X				X
Waterproofness	3.5.5	4.7.4.6	X		X		
Packaging	5.1	4.9	X	X			

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TABLE IV. First article test sequence.

Title	Requirement	Test	First article			
			Preproduction Sample		Initial Preproduction	
			1	2	1	2
Circuit resistance	3.4.1	4.7.3.1	X	X	X	X
Insulation resistance	3.4.2	4.7.3.2	X	X	X	X
Low frequency filter	3.4.3	4.7.3.3	X	X	X	X
High frequency filter	3.4.4	4.7.3.4	X	X	X	X
High temperature test	3.5.1	4.7.4.1	X	X	X	X
Low temperature test	3.5.1	4.7.4.2	X	X	X	X
Vibration	3.5.2	4.7.4.3	X	X	X	X
Shock-basic intensity	3.5.3.1	4.7.4.4.1	X	X	X	X
Shock-high intensity	3.5.3.2	4.7.4.4.2	X	X	X	X
Humidity	3.5.4	4.7.4.5	X		X	
Waterproofness	3.5.5	4.7.4.6	X	X	X	X
Packaging	5.1	4.9				X

TABLE V. Sampling plan for examination.

Inspection lot size	Sample size for examination Major and Minor
2 to 8	5
9 to 15	5
16 to 25	5
26 to 50	5
51 to 90	7
91 to 150	11
151 to 280	13
281 to 500	16
501 to 1200	19
1201 to 3200	23
3,201 to 10,000	29
10,001 to 35,000	35
35,001 to 150,000	40
150,001 to 500,000	40
500,001 and over	40

4.5.2 Examination. The sample selected in accordance with 4.5.1.2 shall be examined and defects classified as specified in table VI (see 4.7.2). The acceptance number in all cases is zero.

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TABLE VI. Classification of defects.

Category	Defect	Method of Inspection
Major		
101	Dimensions affecting interchangeability, not within tolerance (see 3.3).	SIE 1/
102	Improper weight (see 3.3.1).	SIE
103	Protective finish application (see 3.6).	Visual
104	Improper marking (see 3.7).	Visual
105	Improper workmanship affecting interchangeability (see 3.8).	Visual
Minor		
201	Dimensions not affecting interchangeability not within tolerance (see 3.3).	SIE
202	Improper workmanship affecting appearance (see 3.8).	Visual

1/ SIE = Standard inspection equipment.

4.5.2.1 QCI examination failure. Any item that fails to conform to any specified requirement shall be rejected; any failure (one or more) of the selected sample in either the Major/Minor categories for the appropriate inspection lot size shall constitute a failure of the entire lot. The rejected item(s) may be repaired or corrected and resubmitted for inspection. If the contractor utilizes sampling inspection as an element of his inspection system, rejected inspection lots may be resubmitted for acceptance if the contractor performs 100 percent inspection on the lot for those characteristics which were defective and resulted in rejection of the lot and removes all defective units or obtains procuring activity approval to resample the lot due to the insignificance of the defects. Resubmitted lots shall be kept separate from new lots and shall be clearly identified as resubmitted lots.

4.5.3 Tests. Each item shall be subjected to the (QCI) tests specified in table III.

4.5.3.1 QCI test failure. Failure of the item to pass the QCI tests shall be cause for the Government to refuse acceptance of the item. Any item containing one or more defects either shall not be submitted for Government acceptance or shall be corrected and retested and resubmitted without defects for Government acceptance. Resubmitted items shall be kept separate from new items, and shall be clearly identified as resubmitted items.

4.6 Control tests. Control tests shall be conducted on items selected in accordance with 4.6.1 or 4.6.2, as applicable, and subjected to tests specified in table III.

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4.6.1 Frequency (1/50). One assembly shall be selected from each lot of 50 assemblies consecutively produced except that not more than one assembly shall be selected in a one (1) month period, nor less than one test in a three (3) month period.

4.6.2 Frequency (1/200). One assembly shall be selected from each lot, consecutively produced, except that not more than one shall be selected in a two (2) month period, nor less than one test in a six (6) month period.

4.6.3 Failure. Failure of any filter to pass any of the specified control tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.7 Methods of inspection.

4.7.1 Materials, design and construction. Conformance to 3.2, 3.2.1, 3.3, 3.3.2 and 3.3.3 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.7.2 Defects. Conformance to 3.3, 3.3.1, 3.6, 3.7 and 3.8 shall be determined by examination for defects listed in table VI. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.7.3 Performance tests.

4.7.3.1 Circuit resistance. To determine conformance to 3.4.1, apply specified maximum test current to the specified pins indicated in table I. Measure voltage drop across each set of pins and calculate resistance and verify that it conforms to table I.

4.7.3.2 Insulation resistance. To determine conformance to 3.4.2, with a voltage of 50 + 5 vdc applied, measure insulation resistance between pins (except P) of connector P1 or J1, and between pins (except P) and case in accordance with MIL-STD-202, method 302. Verify that the resistance is not less than 10 megohms.

4.7.3.3 Low frequency filter. To determine conformance to 3.4.3, measure capacitance between pin P of connector P1 or J1 and the assembly case in accordance with MIL-STD-202, method 305; the dc bias voltage shall be 2.0 ± 0.2 vdc and alternating current (ac) voltage shall be limited to 1.0 volts root mean square (vrms) at a frequency of 120 ± 5 Hertz (Hz). Verify that capacitance is between 2210 and 4260 microfarads. Measure dc leakage at 72 ± 5°F with a positive voltage of 75 (+0, -5) vdc applied to pin P and verify that leakage current does not exceed 366 microamperes. When dc leakage current is measured during temperature tests of 4.7.4.1 and 4.7.4.2, leakage current shall not exceed 366 microamperes at low temperature (-25°F) and shall not exceed 2.20 milliamperes at high temperature (+125°F).

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4.7.3.4 High frequency filter. To determine conformance to 3.4.4, measure attenuation of each high frequency filter circuit of the assembly in accordance with MIL-STD-220 with the exception that no direct current source be used. During and after this test, verify that the assembly provides specified attenuation at specified frequencies. The characteristic or surge impedance shall be 50 ohms.

4.7.4 Environmental tests.

4.7.4.1 High temperature test. To determine conformance to 3.5.1, subject the assembly to the high temperature test specified in MIL-STD-810, method 501, procedure II, except storage temperature of step 4 shall be 160°F. At conclusion of the storage temperature test, stabilize the assembly at 125°F and subject it to performance tests of 4.7.3.3 and 4.7.3.4. After testing, return the assembly to room ambient temperature and verify that it passes the performance tests of 4.7.3.1 to 4.7.3.4.

4.7.4.2 Low temperature. To determine conformance to 3.5.1, subject the assembly to the low temperature test specified in MIL-STD-810, method 502, procedure I. Maintain temperature at -65°F for a period of not less than 12 hours. At conclusion of the storage temperature test, stabilize the assembly at -25°F and subject it to performance tests of 4.7.3.3 and 4.7.3.4. After testing, return the assembly to room ambient temperature and verify that it passes the performance tests of 4.7.3.1 to 4.7.3.4.

4.7.4.3 Vibration test. To determine conformance to 3.5.2, subject the assembly to the vibration test specified in MIL-STD-810, method 514, procedure VIII, except the vibration level shall be as shown in figure 1. Sweep time shall be 16 minutes per cycle and total cycling time per axis shall be 80 minutes in each of three mutually perpendicular axes. Resonance search shall be conducted at room temperature with the input power supplied. After resonance (if any) has been determined, place the assembly in a 125°F ambient environment and subject it to vibration sweeps with resonance dwell at points (if any) detected. During vibration testing, input power shall be applied and output circuits shall be monitored for discontinuity. At the conclusion of the test, return the assembly to room temperature and verify that it passes the performance tests of 4.7.3.1 to 4.7.3.4.

4.7.4.3.1 Control test (vibration). The test shall be the same as 4.7.4.3 except the dwell time at each resonance shall be two minutes and the test shall be conducted at 73 ± 18°F ambient temperature. At conclusion of the test, verify that the assembly passes the performance tests of 4.7.3.1 to 4.7.3.4.

4.7.4.4 Shock test.

4.7.4.4.1 Basic intensity. To determine conformance to 3.5.3.1, subject the assembly to the shock test specified in MIL-STD-810, method 516, procedure I, for a total of eighteen half-sine wave shock pulses (six in each axis). Peak amplitude shall be 50 ± 5g with a time duration of 18 ± 3 ms measured at 10 percent amplitude points. During the test, apply input power and monitor for intermittent conditions. At the conclusion of the test, subject the assembly to the performance tests of 4.7.3.1 to 4.7.3.4.

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4.7.4.4.1.1 Control test (basic intensity). The test shall be the same as 4.7.4.4.1 except the shock level shall be limited to 37.5 + 3.7g. At the conclusion of the test, verify that the assembly passes the performance tests of 4.7.3.1 to 4.7.3.4.

4.7.4.4.2 High intensity. To determine conformance to 3.5.3.2, subject the assembly to the shock test specified in MIL-STD-810, method 516, procedure IV for a total of eighteen half-sine wave shock pulses. Peak amplitude shall be 125 + 12.5g with a duration of 0.6 + 0.2 ms measured at 10 percent amplitude points. No tests are required during the shock pulses. At the conclusion of the test, verify that the assembly passes the performance tests of 4.7.3.1 to 4.7.3.4.

4.7.4.4.2.1 Control test (high intensity). The test shall be the same as 4.7.4.2 except the shock level shall be limited to 93.7 + 9.3g. At the conclusion of the test, verify that the assembly passes the performance tests of 4.7.3.1 to 4.7.3.4.

4.7.4.5 Humidity. To determine conformance to 3.5.4, place the assembly in a humidity chamber and subject it to the humidity test specified in MIL-STD-810, method 507, procedure II. During and after the test, verify that the assembly passes the performance tests of 4.7.3.1 and 4.7.3.4.

4.7.4.6 Waterproofness. To determine conformance to 3.5.5, the assembly shall be capped at all connections, submerged in water, and a pressure differential of 6 psi developed such that the assembly internal air pressure exceeds external pressure. Pressure differential shall be maintained for a two-minute period without evidence of leakage. After exposure, the caps shall be removed and the assembly subjected to the performance tests of 4.7.3.3 and 4.7.3.4.

4.7.5 Workmanship. Perform visual examinations on each item of the inspection lot at all phases of inspection testing to verify conformance to MIL-STD-454, requirement 9. In addition, subject the assembly to a thermal run-in and a vibration run-in as follows:

a. Thermal run-in. Prior to the acceptance tests of 4.7.3.1 through 4.7.3.4, with the input power applied, verify that the assembly passes temperature cycling per figure 2 for a total cycling time of 24 hours.

b. Vibration run-in. Prior to the performance tests of 4.7.3.1 through 4.7.3.4, with the power off, verify that the assembly passes a vibration cycle having the following characteristics:

1.	Acceleration:	1.0 ± 0.1g peak
2.	Logarithmic frequency sweep:	30 ± 1.0 minute duration from 5 to 500 to 5 Hz
3.	Vibration profile:	As specified in figure 3

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4.9 Inspection of packaging. The assembly shall be inspected by the supplier prior to shipment to ensure conformance to preservation, packaging, packing and marking requirements in Section 5. Sampling inspections shall be limited to those characteristics that can be determined by visual examination.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The EMI filter assembly is intended for use primarily to provide high and low frequency filtration in the electrohydraulic stabilization system of the M60A1 tank.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification, and military part number.
b. Whether first article is required (see 3.1).
c. If test conditions should be other than as specified (see 4.3).
d. Arrangements for first article inspection and rights of the Government (see 4.4.1, 4.4.2 and 6.3).
e. If sampling plan for QCI is other than as specified (see 4.5.1).
f. Selection of applicable level and packaging requirements (see 5.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.4 Definitions.

6.4.1 Definitions of terms used in sampling inspection.

a. Classification of defects. A classification of defects is the enumeration of possible defects of the unit of product classified according to their seriousness. A defect is any nonconformance of the unit of product with specified requirements. Defects will normally be grouped into one or more of the following classes: critical, major and minor defects. Also, defects may be grouped into other classes, or into subclasses within these classes.

b. Critical defects. A critical defect is a defect that judgement and experience indicate would result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product, or a defect that judgement and experience indicate is likely to prevent performance of the tactical function of a major end item such as a ship, aircraft, tank, missile, or space vehicle.

c. Critical defective. A critical defective is a unit of product which contains one or more critical defects and may also contain major and/or minor defects.

d. Defective. A defective is a unit of product which contains one or more defects.

e. Formation of lots or batches. The product shall be assembled into identifiable lots, sublots, batches, or in such other manner as may be prescribed (see 1). Each lot or batch shall, as far as is practicable, consist of units of product of a single type, grade, class, size, and composition, manufactured under essentially the same conditions, and at essentially the same time.

f. Lot or batch. The term lot or batch shall mean "inspection lot" or "inspection batch", i.e., a collection of units or product from which a sample is to be drawn and inspected and may differ from a collection of units designated as a lot or batch for other purposes (e.g., production, shipment, etc.).

g. Lot or batch size. The lot or batch size is the number of units of product in a lot or batch.

h. Major defect. A major defect is a defect, other than critical, that is likely to result in failure, or to reduce materially the usability of the unit of product for its intended purpose.

i. Major defective. A major defective is a unit of product which contains one or more major defects, and may also contain minor defects but contains no critical defect.

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j. Minor defect. A minor defect is a defect that is not likely to reduce materially the usability of the unit of product for its intended purpose, or is a departure from established standards having little bearing on the effective use or operation of the unit.

k. Minor defective. A minor defective is a unit of product which contains one or more minor defects but contains no critical or major defect.

l. Presentation of lots or batches. The formation of the lots or batches, lot or batch size, and the manner in which each lot or batch is to be presented and identified by the supplier shall be designated or approved by the responsible authority. As necessary, the supplier shall provide adequate and suitable storage space for each lot or batch, equipment needed for proper identification and presentation, and personnel for all handling of product required for drawing of samples.

m. Representative sampling. When appropriate, the number of units in the sample shall be selected in proportion to the size of sublots or subbatches, or parts of the lot or batch, identified by some rational criterion. When representative sampling is used, the units from each part of the lot or batch shall be selected at random.

n. Sample A sample consists of one or more units of product drawn from a lot or batch, the units of the sample being selected at random without regard to their quality. The number of units or product in the sample is the sample size.

o. Sampling plan. A sampling plan indicates the number of units of product from each lot or batch which are to be inspected (sample size or series of sample sizes) and the criteria for determining the acceptability of the lot or batch (acceptance and rejection numbers).

p. Time of sampling. Samples may be drawn after all the units comprising the lot or batch have been assembled, or samples may be drawn during assembly of the lot or batch.

6.4.2 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.4.3).

6.4.3 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows, or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

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6.5 Subject term (key word) listing.

Circuit resistance
Insulation resistance
Low frequency
High frequency
Temperature resistance
Vibration resistance
Shock resistance
Waterproofness

6.6 AMC policy on AQLs/LTPDs. This specification is certified to be in compliance with current Army Materiel Command (AMC) policy for the elimination of AQLs/LTPDs (Acceptable Quality Levels/Lot Tolerance Percent Defectives) from military specifications.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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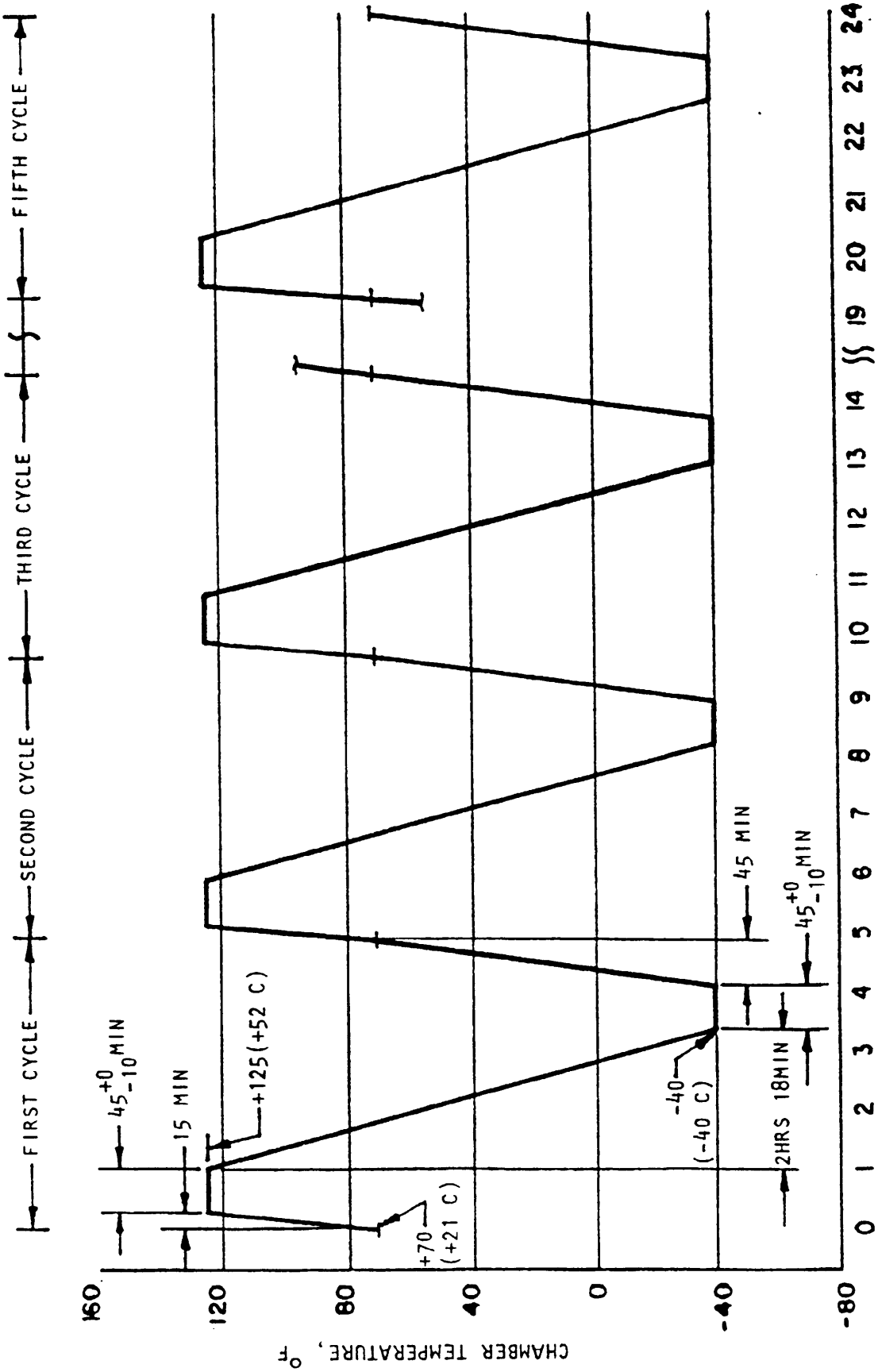


FIGURE 2. Temperature cycling.

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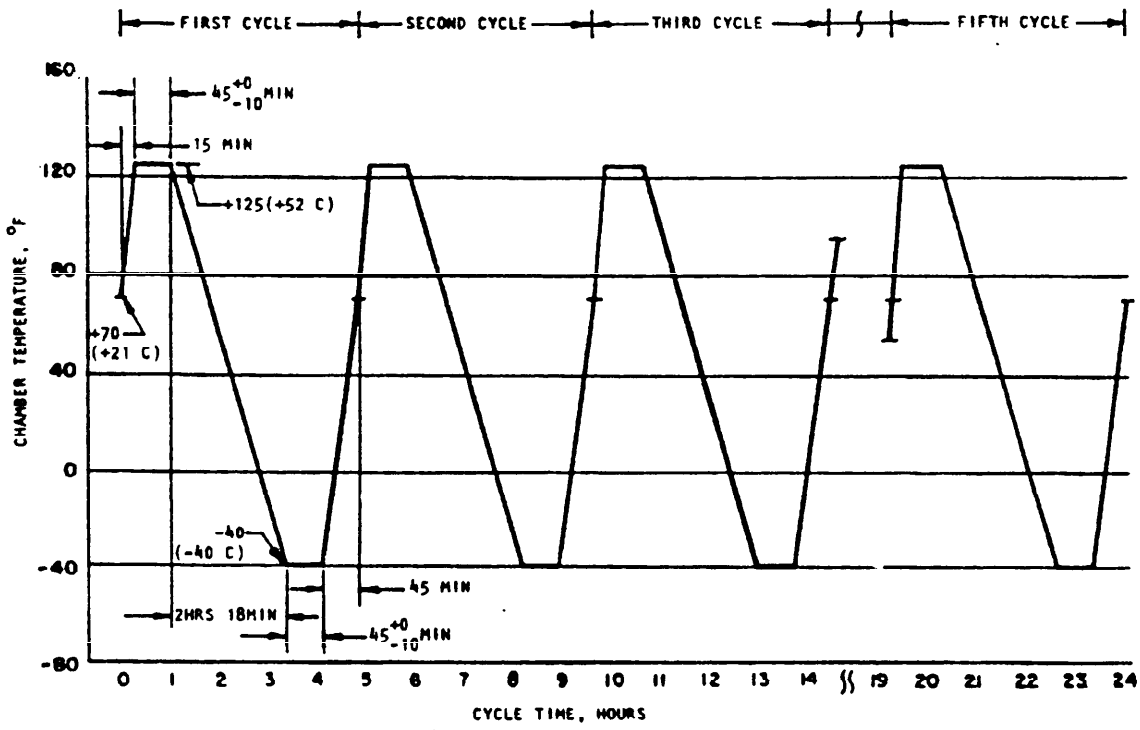
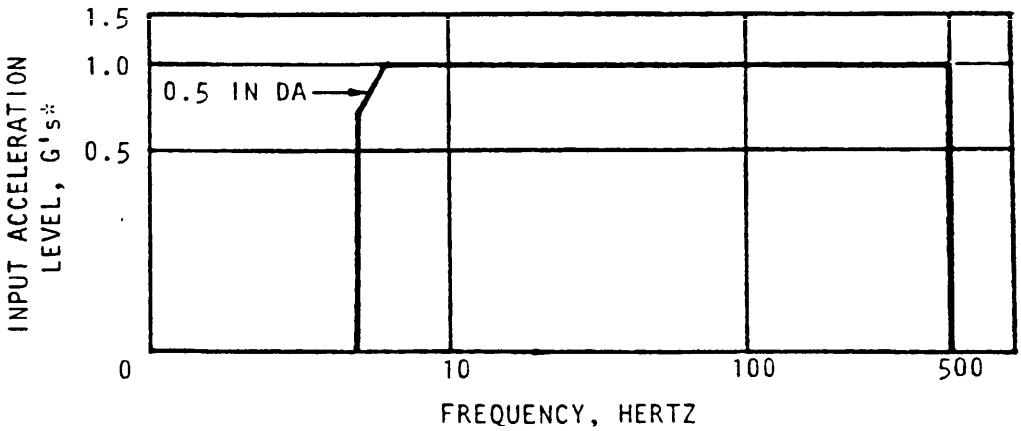


FIGURE 2. Temperature cycling.



*APPLIED AS AN INPUT TO THE COMPONENT MOUNTING BRACKET

FIGURE 3. Vibration profile.

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Preparing activity:
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